

BUZAU, Romania

WMO: 153500

Lat: 45.1328N

Lon: 26.8517E

Elev: 98

StdP: 100.15

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.1	-9.5	-15.9	0.9	-10.4	-13.2	1.2	-8.1	16.3	-1.6	11.9	-3.6	3.4	50	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	33.8	21.3	32.2	20.8	30.7	20.4	23.6	30.0	22.5	28.8	21.6	28.0	3.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.7	16.5	27.1	20.4	15.3	25.7	19.4	14.4	24.6	71.1	30.4	66.7	28.9	63.2	27.9	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	8.9	7.6	-16.3	36.4	3.4	1.8	-18.8	37.6	-20.7	38.7	-22.6	39.7	-25.0	41.0		
			-16.7	24.8	3.3	1.7	-19.0	26.0	-21.0	27.0	-22.8	28.0	-25.2	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-1.0	1.6	6.4	12.2	17.8	21.7	23.8	23.7	18.5	12.2	6.4	0.7
	DBStd	9.58	4.70	5.15	4.52	3.76	3.44	3.10	2.77	3.06	3.57	4.10	4.25	4.52
	HDD10.0	1159	340	236	127	21	1	0	0	0	0	25	122	289
	HDD18.3	2821	598	468	369	186	52	8	1	2	42	192	357	547
	CDD10.0	1912	0	1	16	87	242	350	428	424	254	94	15	0
	CDD18.3	532	0	0	0	2	35	108	170	168	46	3	0	0
	CDH23.3	4891	0	0	1	34	324	953	1591	1592	364	32	0	0
	CDH26.7	1835	0	0	0	5	71	322	654	681	99	3	0	0
Wind	WSAvg	3.2	3.3	3.5	3.9	3.6	3.2	2.9	2.9	2.9	3.0	2.9	3.0	3.2
Precipitation	PrecAvg	537	25	19	28	42	67	77	67	52	50	45	32	33
	PrecMax	763	60	56	61	110	195	117	170	129	117	131	76	89
	PrecMin	342	2	2	0	17	19	23	7	10	4	3	1	1
	PrecStd	112	16	15	16	21	41	25	42	34	30	30	22	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	17.0	22.0	27.2	30.9	33.9	36.2	36.6	32.1	26.8	20.5	13.5
		MCWB	7.6	10.9	12.0	16.3	19.4	21.6	20.8	21.1	19.3	17.3	14.2	9.5
	2%	DB	9.6	14.0	19.1	23.6	28.5	31.7	33.8	34.1	29.4	23.7	17.2	10.5
		MCWB	5.8	8.9	11.1	14.9	18.3	21.1	21.2	21.5	18.7	16.3	12.7	7.9
	5%	DB	6.9	11.1	16.5	21.3	26.8	30.1	32.1	32.3	27.3	21.1	14.6	8.3
		MCWB	4.8	7.3	9.8	13.5	17.4	20.6	21.1	20.5	17.7	14.9	11.2	6.1
	10%	DB	5.1	8.4	14.0	19.2	25.0	28.5	30.4	30.6	25.3	18.7	12.5	6.4
		MCWB	3.6	5.7	8.4	12.4	16.6	19.8	20.8	20.1	17.0	13.8	10.2	4.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	11.7	13.8	17.2	21.5	24.0	25.1	26.0	22.5	19.0	15.0	10.4
		MCDB	12.1	15.5	19.5	25.2	29.2	30.4	30.5	31.8	28.6	24.0	19.2	12.5
	2%	WB	6.5	9.4	11.6	15.6	19.7	22.5	23.6	23.8	20.2	17.1	13.4	8.2
		MCDB	8.6	13.1	17.6	22.2	25.9	29.3	29.8	30.1	25.5	21.3	16.7	10.0
	5%	WB	5.2	7.6	10.3	14.4	18.5	21.5	22.6	22.2	18.8	15.6	11.9	6.5
		MCDB	6.8	10.4	15.8	19.9	24.7	28.1	28.8	28.3	24.9	20.1	13.9	7.9
	10%	WB	3.6	6.0	8.9	13.2	17.4	20.6	21.6	21.0	17.8	14.3	10.5	4.9
		MCDB	5.0	8.3	13.1	17.9	23.3	26.7	27.9	27.8	23.4	18.0	12.3	6.2
Mean Daily Temperature Range	5% DB	MDBR	6.5	7.7	9.4	10.4	11.0	10.8	10.8	11.2	10.3	9.1	7.4	6.4
		MCDBR	9.6	12.6	14.7	14.1	13.6	13.0	13.4	13.5	13.3	12.9	11.6	9.6
		MCWBR	6.7	8.2	8.2	7.1	6.0	5.1	4.7	4.6	5.2	6.4	7.3	6.8
	5% WB	MCDBR	9.0	11.8	13.4	12.5	12.2	11.9	11.4	11.8	11.8	11.8	10.2	8.8
		MCWBR	6.6	8.1	8.0	6.7	6.0	5.4	5.2	5.4	5.5	6.5	6.7	6.7
Clear-Sky Solar Irradiance	taub	0.333	0.363	0.398	0.455	0.430	0.433	0.447	0.461	0.422	0.385	0.358	0.335	
	taud	2.332	2.248	2.205	2.061	2.216	2.241	2.207	2.156	2.230	2.315	2.345	2.343	
	Ebn at Noon	761	802	825	805	842	839	821	791	790	768	723	715	
	Edh at Noon	80	104	124	156	138	135	139	141	120	96	79	72	
All-Sky Solar Radiation	RadAvg	1.31	2.14	3.31	4.55	5.73	6.11	6.21	5.59	4.06	2.63	1.44	1.10	
	RadStd	0.18	0.28	0.35	0.47	0.44	0.44	0.43	0.35	0.34	0.30	0.15	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (58 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	-143	+77	+41

Nomenclature: See separate page